

Practice Quiz 10B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Images and inverse images; the geometric sum formula.

1. (a) Let $A = \{0, 1, 2, 3\}$, $B = \{u, v\}$, and define

$$f(0) = u, \quad f(1) = v, \quad f(2) = u, \quad f(3) = v.$$

For $X = \{0, 1\}$, compute $f(X)$ and $f^{-1}(f(X))$. Are the latter and X equal?

Solution $f(X) = \{u, v\} = B$. Every element of A maps into B , so

$$f^{-1}(f(X)) = \{0, 1, 2, 3\} \neq \{0, 1\} = X.$$

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- (b) For real $r \neq 1$, state and prove by induction a closed formula for

$$2 + 2r + 2r^2 + \cdots + 2r^n \quad (n \geq 0).$$

Solution The formula is

$$\sum_{i=0}^n 2r^i = 2 \frac{r^{n+1} - 1}{r - 1}.$$

For $n = 0$, both sides equal 2. Suppose the formula holds for n . Then

$$\begin{aligned} \sum_{i=0}^{n+1} 2r^i &= 2 \frac{r^{n+1} - 1}{r - 1} + 2r^{n+1} \\ &= 2 \frac{r^{n+1} - 1 + r^{n+1}(r - 1)}{r - 1} = 2 \frac{r^{n+2} - 1}{r - 1}. \end{aligned}$$

This is the formula for $n + 1$, so induction proves the result. ■